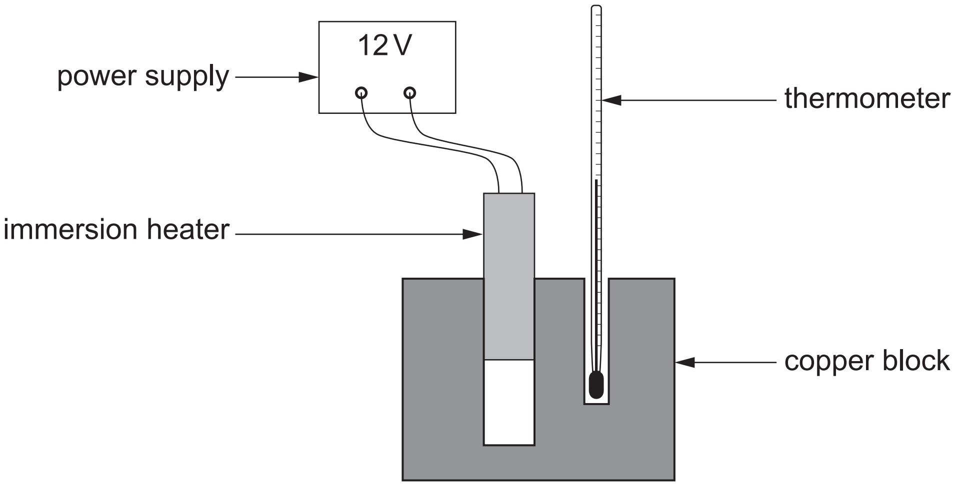
Examiner only

4. The apparatus used by a group of students to determine the specific heat capacity of copper is shown below.



(a) (i) The results from the experiment are given below.

Mass of the copper block = 0.5 kg
Temperature of the copper block at the start of the experiment = 19.5 °C
The energy supplied to the heater during the experiment = 5400 J
The temperature of the copper block after heating = 42.0 °C

Use an equation from page 2 to calculate the specific heat capacity of copper. [3]

specific heat capacity = ... J/kg °C

(ii) The true value of the specific heat capacity of copper is 385 J/kg °C.
Comment on the accuracy of the group's value. [2]

....

....

....

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(iii) State **two** improvements that could be made to this experiment that would lead to a more accurate value for the specific heat capacity of copper. [2]

1. ....

2. ....

(b) Explain what happens to the copper atoms when the block is heated. [2]

....

....

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